A Report

On

SIEMENS DATA SCIENCE MASTER VIRTUAL INTERNSHIP

(AICTE – EduSkills Virtual Internship Program, supported by Siemens)

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE

By

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Name of Student: PRACHI RATHOUR
Batch:

2024-2028

CANDIDATE'S DECLARATION

I, Prachi Rathour, Roll Number 2410030986, do hereby declare that the internship report titled “Siemens Data Science Master Virtual Internship” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own work and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature: _____

Student Name: Prachi Rathour

Roll No.: 2410030986

ACKNOWLEDGEMENT

I'd like to thank everyone who made this internship possible and, honestly, manageable alongside my regular semester workload. Thanks to AICTE and EduSkills for setting up the program in the first place, and to Siemens Industry Software (India) Pvt. Ltd. for backing it. I'm also grateful to the faculty at IILM University, Greater Noida, for their guidance and for being patient with my questions along the way.

The RapidMiner Academy content itself deserves a mention too — the week-wise structure made a genuinely large amount of material feel manageable in eight weeks, which I didn't expect going in.

And finally, thanks to my family and friends for the usual encouragement (and for not asking too many questions about what “attrition prediction” means).

Signature: _____
Student Name: Prachi Rathour
Roll No.: 2410030986

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CHAPTER 3: INTERNSHIP COMPLETION CERTIFICATE



Fig 3.1 — Certificate of Virtual Internship (Data Science Master), AICTE–EduSkills, supported by Siemens

CHAPTER 4: PROJECT DESCRIPTION

4.1 Introduction

Between April and June 2026, I completed the Siemens Data Science Master Virtual Internship, an 8-week program run jointly by AICTE and EduSkills on the RapidMiner AI Hub platform. I'd used RapidMiner briefly in a college lab session before, but this was the first time I worked through a full, graded curriculum on it — from framing a business problem all the way to scoring a model in real time. The program was supported by Siemens Industry Software (India) Pvt. Ltd., and ran under the National Internship Portal of the Ministry of Education.

Each week paired a set of concepts with a hands-on module and a short assessment, so the pace stayed fairly steady — I'd watch the lesson videos over a couple of evenings, then spend the weekend actually building the process in RapidMiner Studio. By the final week I'd sat six certification exams (one Professional-level and five Master-level) and appeared for the Final Assessment Test, which together gave me an overall grade of O (Outstanding).

4.2 Organization Profile

Siemens Industry Software is the digital-industries software arm of Siemens, best known for PLM and industrial automation tools; here, it sponsored the internship as part of a broader skilling push. The program itself was run by EduSkills, a not-for-profit that partners with companies like Siemens to deliver AICTE-recognised virtual internships to engineering students across India — their tagline, “Nation Building through Skills,” shows up on every certificate.

The actual platform I learned on was RapidMiner, now owned by Altair. It's built around three pieces that matter for this report: RapidMiner Studio, where you visually drag-and-drop operators to build a workflow; the AI Hub, which is where finished workflows get stored, scheduled and shared; and Radoop, which lets those workflows run against big-data clusters when a dataset is too large for a single machine.

4.3 Problem Statement

Most of what I'd learned about machine learning in class was fairly theoretical — I could explain how a random forest works or derive gradient descent, but I'd never

actually taken a model from “we have some data” to “here's a live scoring endpoint.” That gap is really what this internship was meant to close: learning to spot which business problems are even worth solving with ML, cleaning and engineering the data behind them, building and tuning models properly, and then dealing with the less glamorous but equally important part — deploying and administering those models on an actual platform.

4.4 Project Objectives

- Get comfortable applying CRISP-DM to a business problem instead of just knowing the diagram.
- Practice real data engineering — loading, cleaning, transforming and calculating fields — inside RapidMiner Studio.
- Build and tune classification, regression, clustering and association-rule models, and know when to use each.
- Pick up the more advanced techniques: proper cross-validation, parameter optimisation, feature engineering, and basic time-series forecasting.
- Understand how a model actually gets deployed — running processes, publishing a web service, and keeping an eye on it afterwards.
- Get hands-on with AI Hub administration itself, including Radoop and Real-Time Scoring configuration.

4.5 Scope of the Project

The internship was self-paced and entirely virtual, using the RapidMiner Studio installation and AI Hub sandbox provided through the program — there was no live client or production system involved, and grading was done module-by-module through certification exams rather than a single end-to-end deliverable. Even so, several of the modules asked us to work through a complete mini-project rather than isolated exercises, and I've summarised one of mine below since it's a fair representation of the kind of work the program covered.

4.5.1 Sample Project: Employee Attrition Prediction

For the Applications & Use Cases and Machine Learning modules, I worked with a sample HR dataset (roughly 4,000 employee records with fields like age, department,

monthly income, years at company, overtime status and job satisfaction score) to predict which employees were likely to leave. After the usual cleanup — handling a handful of missing income values and encoding categorical fields like department and job role — I tried three algorithms inside RapidMiner Studio: Naive Bayes, Decision Tree, and Random Forest, comparing them using 10-fold cross-validation.

Random Forest came out ahead with roughly 87% accuracy and an AUC around 0.84, noticeably better than the Decision Tree's ~79% and Naive Bayes' ~74%. Overtime status and monthly income turned out to be the strongest predictors, which matched what I'd expect intuitively. As a last step for the Applications & Use Cases Master module, I published the trained model through Real-Time Scoring and tested it by sending a few sample employee records to the web service and checking the returned predictions manually.

4.6 Technologies and Tools Used

Tool / Technology	Purpose
RapidMiner Studio	Visual, no-code/low-code environment for building end-to-end data science workflows.
RapidMiner AI Hub	Central platform for managing, sharing, scheduling and deploying models and processes.
RapidMiner Radoop	Enables execution of RapidMiner processes on big-data clusters (Hadoop/Spark).
Real-Time Scoring	Exposes trained models as web services for real-time predictions.
RapidMiner Marketplace	Extends functionality through community and vendor-built extensions.
CRISP-DM Methodology	Standard process model followed for structuring the data science workflow.
R / Python Integration	Used to extend built-in operators with custom scripts where required.

4.7 System Architecture

It helps to think of the RapidMiner setup as a fairly standard client–server split. Studio is the client — it's where you actually build a “process” by chaining operators together for reading data, transforming it, training a model and evaluating it. Once a process works the way you want, you publish it to the AI Hub, which is the server

side: it stores the process, lets you schedule it to run on its own, and keeps track of who has access to what.

When the data is too big for a single machine — which wasn't really the case for my own datasets, but is covered in the Data Engineering Master module — Radoop pushes the same kind of process down onto a Hadoop or Spark cluster instead. And once a model has been trained and validated, Real-Time Scoring turns it into a REST endpoint, so any external application can send it a record and get a prediction back instantly. The Marketplace sits on top of all this, adding community- or vendor-built operators when the built-in ones aren't enough.

4.8 Methodology

The program followed a fixed week-wise plan — a new module every week, a short assessment at the end of it, and for a few modules, an actual assignment to submit. Everything came together in the final week with a Final Assessment Test, and the grade I received combined all of it: the weekly quizzes, the submitted work, and that final test. The table below is basically the syllabus as it was laid out in my offer letter.

Week	Module	Description
Week 1	Applications & Use Cases Professional	Learn to identify business problems suitable for AI and machine learning.
Week 2	Data Engineering Professional	Learn to access, load, transform and calculate data.
Week 3	Machine Learning Professional	Build foundational machine learning skills.
Week 4	Data Preparation Master	Master advanced data handling and automation.
Week 5	Machine Learning Master	Learn advanced machine learning techniques.
Week 6	Applications & Use Cases Master	Deploy and monitor ML models.
Week 7	Platform Administration Master	Configure and manage RapidMiner AI Hub.
Week 8	Platform Administration Master	Enterprise-level platform operations.

Underneath the week-by-week structure, the modules still followed CRISP-DM as the overall framework — business understanding, data understanding, data preparation,

modelling, evaluation, deployment — which made it easier to see how a given week's topic fit into the bigger picture rather than feeling like a standalone lesson.

4.9 Expected Outcomes

- Six RapidMiner certifications — one Professional and five Master level — across Applications & Use Cases, Machine Learning, Data Engineering and Platform Administration.
- Enough hands-on practice to build, validate and deploy a full ML workflow myself, rather than just describing the steps on paper.
- A much better sense of what "production" actually involves — scoring endpoints, scheduling, access control — which I think will matter for placements more than another theory course would have.
- A working sample project (the attrition model in 4.5.1) that I can point to in interviews, plus the underlying processes I can reuse or extend later.

4.10 Certificates of Completion and Other Communication Proof

The offer letter and all six certificates are attached on the following pages as proof of completion.

Internship Offer Letter

EduSkills®
Nation Building Through Skills

Ref No: C16Y2026M081441027

Internship Offer Letter

Student Name	: Prachi Rathour
Stream	: B.Tech
Branch	: B.Tech(Computer Science and Engineering)
Institute Name	: IILM University, Greater Noida
Internship Domain	: Siemens Data Science Master
Internship Duration	: April 2026 - June 2026
AICTE Student ID	: STU6836a9321c16f1748412722

We are pleased to inform you that you have been selected for the 8-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

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EduSkills®
Nation Building Through Skills

The week wise Structure of the Internship is as Follows:

Week	Module	Description
Week 1	Applications & Use Cases Professional	Learn to identify business problems suitable for AI and machine learning.
Week 2	Data Engineering Professional	Learn to access, load, transform, and calculate data.
Week 3	Machine Learning Professional	Build foundational machine learning skills.
Week 4	Data Preparation Master	Master advanced data handling and automation.
Week 5	Machine Learning Master	Learn advanced machine learning techniques.
Week 6	Applications & Use Cases Master	Deploy and monitor ML models.
Week 7	Platform Administration Master	Configure and manage RapidMiner AI Hub.
Week 8	Platform Administration Master	Enterprise-level platform operations.
Final Credential Validation / Internship Grade Point Assessment		

Note: There is no stipend associated with this internship.

We congratulate you on your selection and wish you a successful learning journey.

Bibek Ranjan
Director, EduSkills

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Fig 4.10(a) — AICTE–EduSkills Internship Offer Letter

RapidMiner Certifications



Fig 4.10(b) — Applications & Use Cases Professional Certification



Fig 4.10(c) — Machine Learning Professional Certification



Fig 4.10(d) — Data Engineering Master Certification



Fig 4.10(e) — Machine Learning Master Certification



Fig 4.10(f) — Applications & Use Cases Master Certification



Fig 4.10(g) — Platform Administration Master Certification

CHAPTER 5: BIBLIOGRAPHY / REFERENCES

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